



Data Science - Project Course

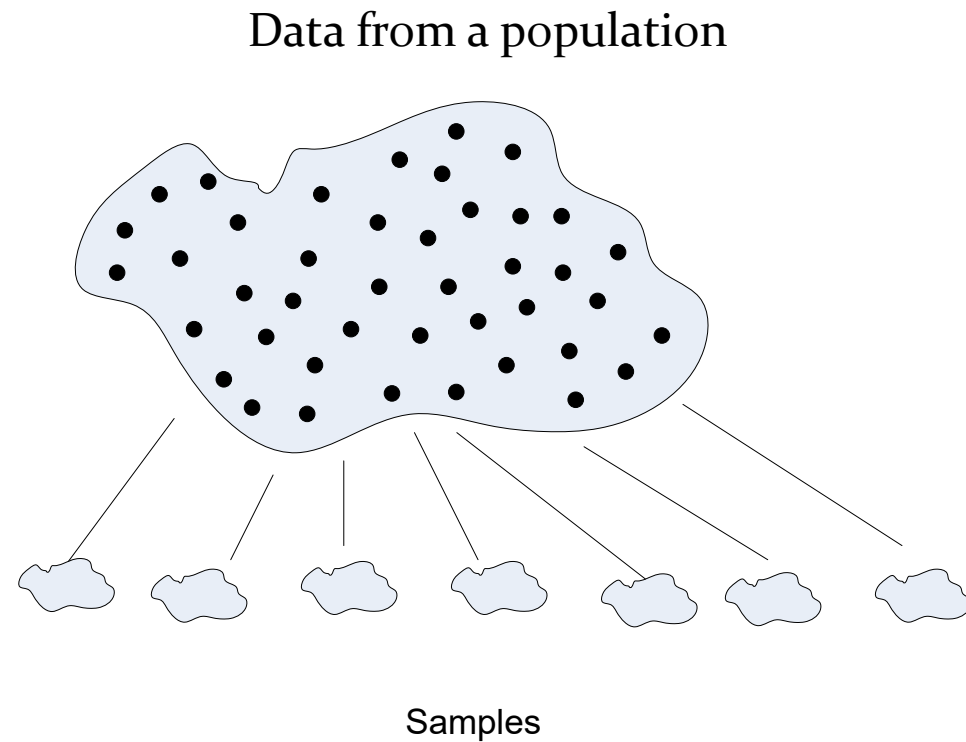
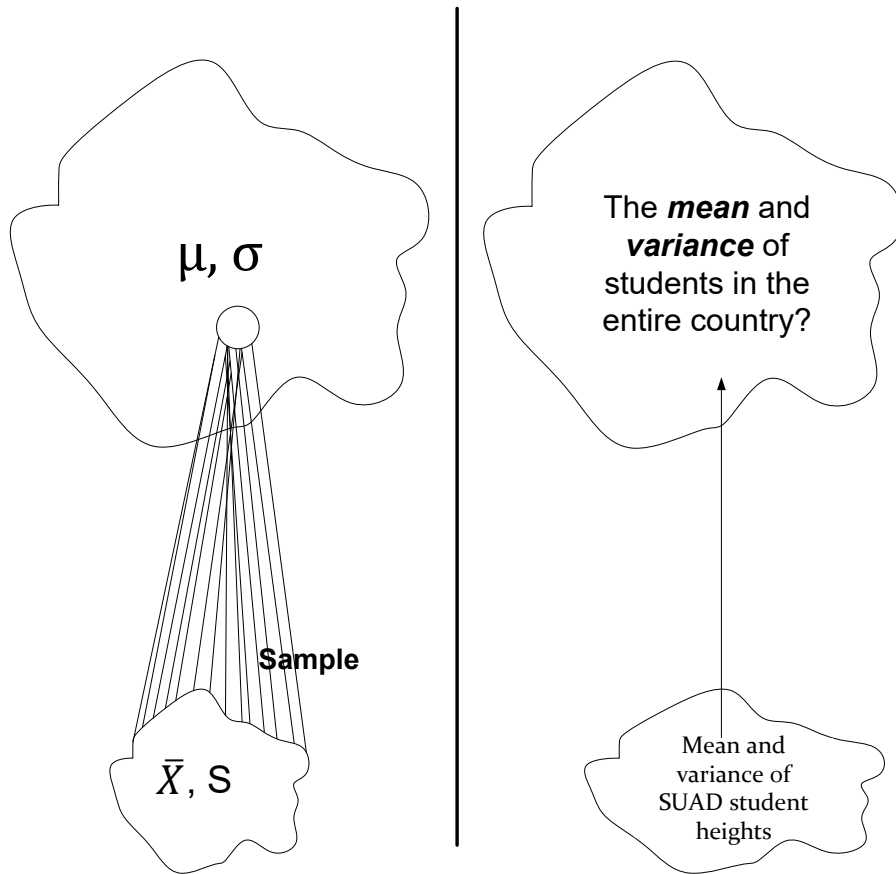
Lecture 3: Correlation Analysis

Dr. Tanujit Chakraborty

Faculty @ Sorbonne

tanujitisi@gmail.com

Quick Recap...





Quick Recap...

The primary objective of statistical analysis is to use data from a sample to make inferences about the population statistics from which the sample was drawn.

- Our main purpose in selecting random samples is to elicit information about the unknown population parameters.
- Any function of the random variables constituting a random sample is called a **statistic**. It's probability distribution is called **Sampling Distribution** (e.g., Chi-square, t, F distributions).

Quick Recap...

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^N x_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$
N = number of items in the population	n = number of items in the sample

Population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{\sigma^2}$$

Vs

Sample

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$S = \sqrt{S^2} \quad \text{Why?}$$

Unbiasedness of Sample Mean

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^N x_i}{N}$ N = number of items in the population	$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$ n = number of items in the sample

$$\begin{aligned}
 E(\bar{X}) &= \frac{1}{n} [E(X_1 + X_2 + \dots + X_n)] \\
 &= n \times \frac{1}{n} \mu \\
 &= \mu
 \end{aligned}
 \qquad
 \begin{aligned}
 Var(\bar{X}) &= \frac{1}{n^2} [Var(X_1 + X_2 + \dots + X_n)] \\
 &= n \times \frac{1}{n^2} \sigma^2 \\
 &= \frac{\sigma^2}{n}
 \end{aligned}$$

Unbiasedness of Sample Variance

Given a distribution with variance σ^2 , the sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator for σ^2 .

$$E(S^2) = E \left[\frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2) \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n (V(X_i) + [E(X_i)]^2) - n(V(\bar{X}) + [E(\bar{X})]^2) \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n (\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right) \right] = \sigma^2$$

Population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{\sigma^2}$$

Vs

Sample

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$S = \sqrt{S^2} \quad \text{Why?}$$

This presentation includes...

- Introduction to Relationship Analysis
- Correlation Analyses
- Measures of Correlations
 - Karl Pearson's correlation coefficient
 - Charles Spearman's correlation coefficient
 - Chi-square coefficient of correlation
- Spurious Correlation

Data for Relationship Analysis

Univariate population: The population consisting of only one variable.

Example:

<i>Temperature</i>	20	30	21	18	23	45	52
--------------------	----	----	----	----	----	----	----

Here, statistical measures suffice to find a relationship.

Bivariate population: Here, the data happen to be with two variables.

Example:

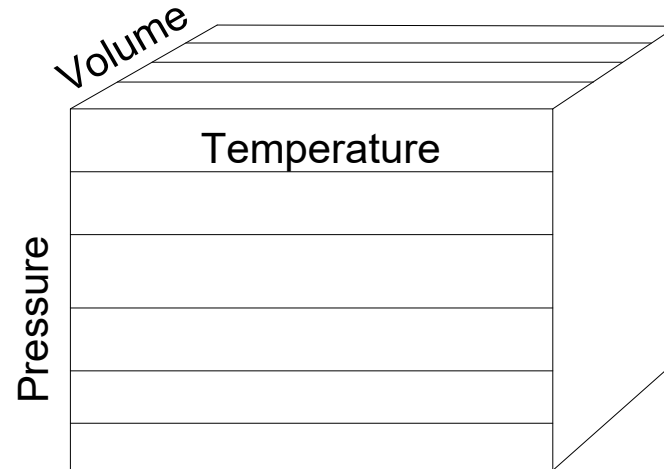
<i>Pressure</i>	1	1.1		0.8
<i>Temperature</i>	35	41		29

Data for Relationship Analysis



Multivariate population: If the data happen to be one more than two variable.

Example:



If we add another variable say viscosity in addition to Pressure, Volume or Temperature?

Measures of Relationship



In case of bivariate and multivariate populations, usually, we have to answer two types of questions:

Q1: Does there exist relation between two variables (in case of bivariate population) ?

- If yes, of what degree?

Q2: Is there any relationship between one variable in one side and two or more variables on the other side (in case of multivariate population)?

- If yes, of what degree and in which direction?

Measures of Relationship

In case of bivariate and multivariate populations, usually, we have to answer two types of questions:

Q1: Does there exist relation between two variables (in case of bivariate population) ?

Q2: Is there any relationship between one variable in one side and two or more variables on the other side (in case of multivariate population)?

Solution

?

Measures of Relationship

In case of bivariate and multivariate populations, usually, we have to answer two types of questions:

Q1: Does there exist relation between two variables (in case of bivariate population) ?

Q2: Is there any relationship between one variable in one side and two or more variables on the other side (in case of multivariate population)?

To find solutions to the above questions, two approaches are known.

**Correlation
Analysis**

**Regression
Analysis**



Correlation Analysis



Correlation Analysis

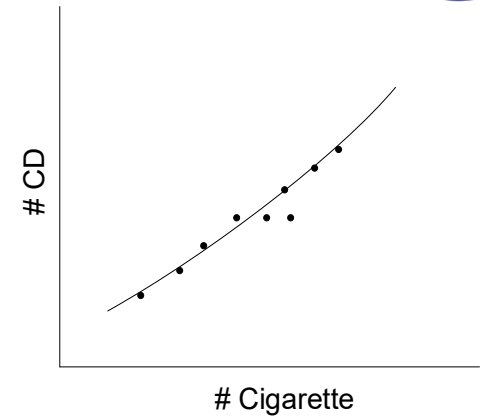
In statistics, the word correlation is used to denote some form of association between two variables.

Example: Weight is correlated with height

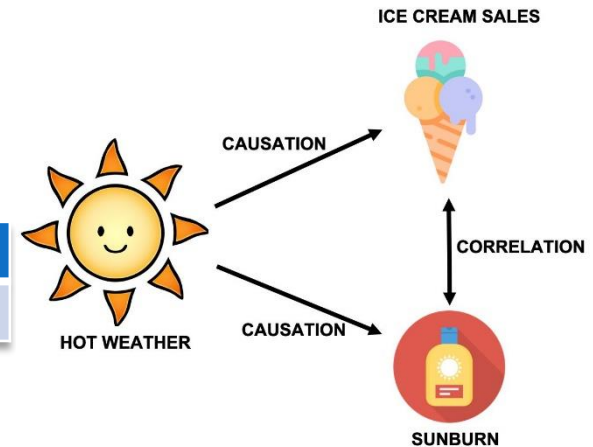
Correlation Analysis

Do you find any correlation between X and Y as shown in the table?

Look at the table given below



<i>No. of CD's sold in shop X</i>	25	30	35	42	48	52	56
<i>No. of cigarette sold in Y</i>	5	7	9	10	11	11	12



Note

- In data analytics, a correlation analysis makes sense only when a relationship makes sense.
- Correlation does NOT imply causation.

Correlation Analysis

A	a_1	a_2	a_3	a_4	a_5	a_6	
B	b_1	b_2	b_3	b_4	b_5	b_6	

Correlation

Positive correlation

If the value of the attribute **A** **increases** with the **increase** in the value of the attribute **B** and vice-versa.

Negative correlation

If the value of the attribute **A** **decreases** with the **increase** in the value of the attribute **B** and vice-versa.

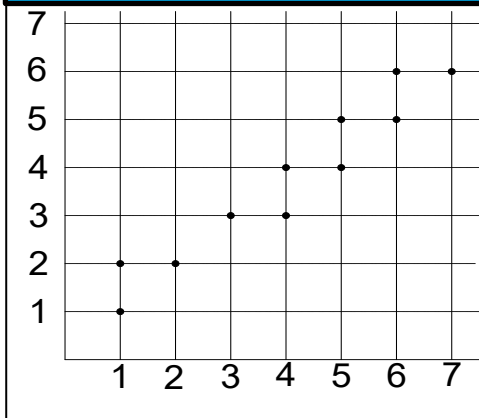
Zero correlation

When the values of attribute **A** varies **at random** with **B** and vice-versa.

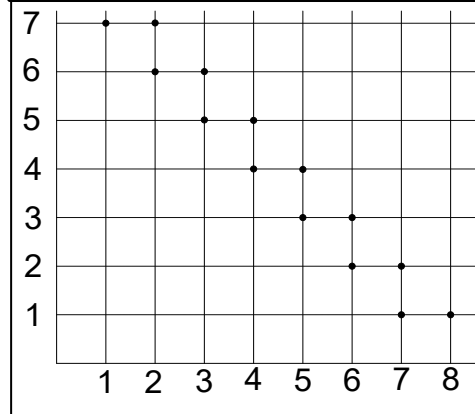
Correlation Analysis



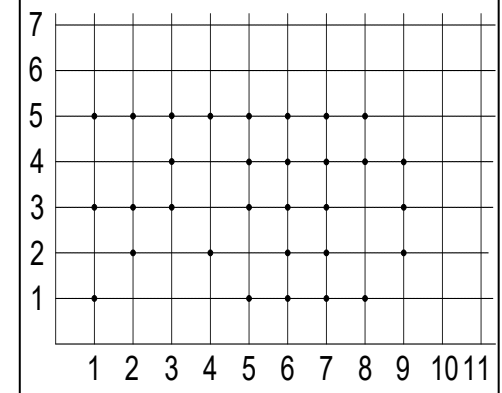
Positive correlation



Negative correlation



Zero correlation

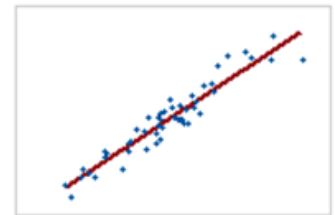


Form of Correlation

Concerning the form of a correlation, it could be linear, non-linear, or monotonic.

Linear Correlation

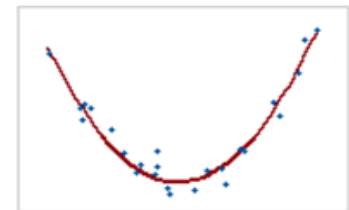
A correlation is linear when two variables change at constant rate.



Linear Correlation

Non-linear Correlation

In this case, the relationship between the variables graph as a curved pattern (parabola, hyperbola ... etc).



Non-linear Correlation

Form of Correlation

Concerning the form of a correlation, it could be linear, non-linear, or **monotonic**.

Monotonicity of a function

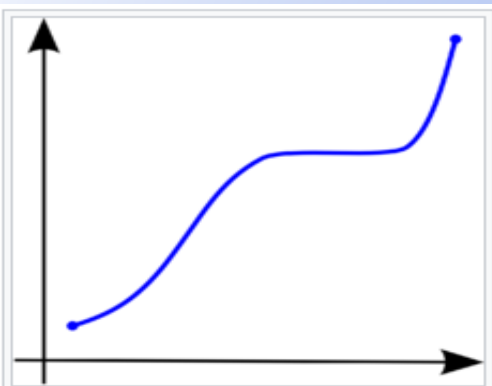


Figure 1. A monotonically increasing function.

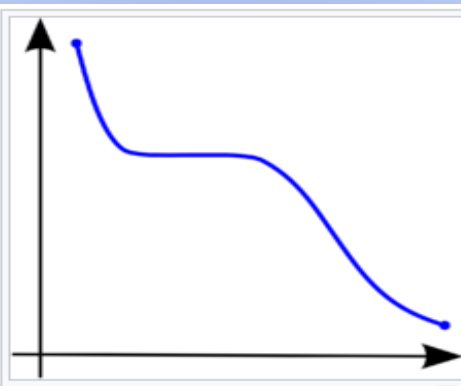


Figure 2. A monotonically decreasing function

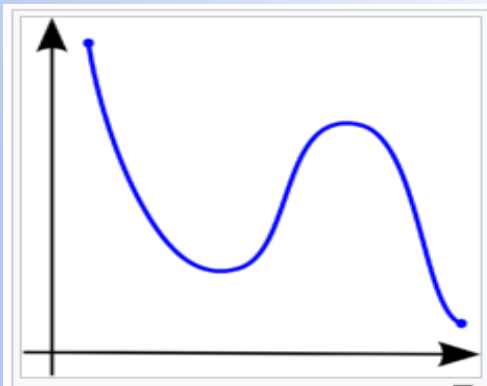


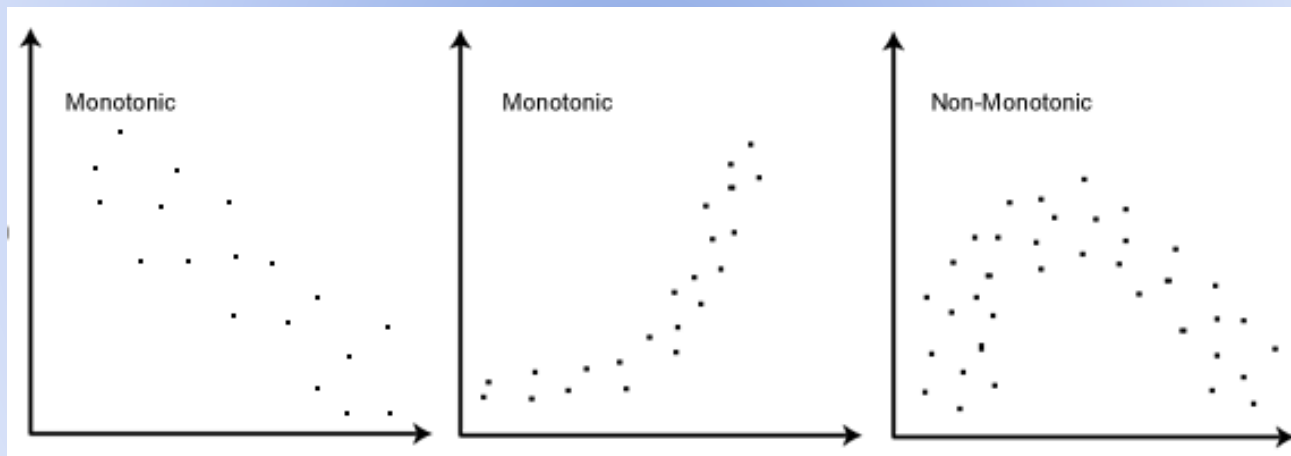
Figure 3. A function that is not monotonic

Form of Correlation

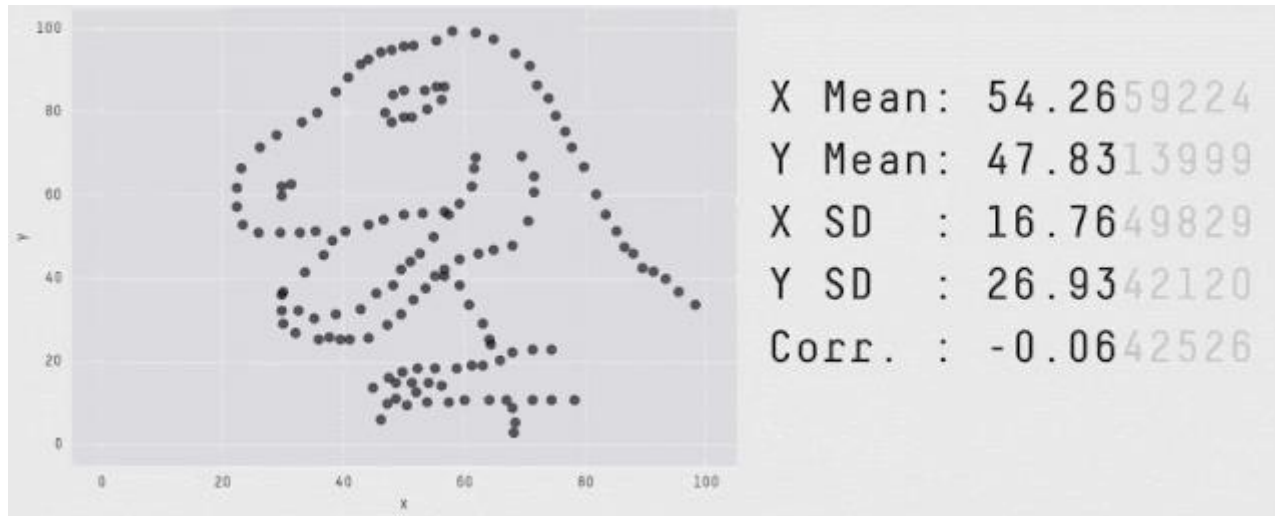
Concerning the form of a correlation, it could be linear, non-linear, or **monotonic** :

Monotonic and non-monotonic relations

Monotonic correlation: In a monotonic relationship, the variables tend to move in the same relative direction or opposite direction, but not necessarily at a constant rate.



Spurious Correlation



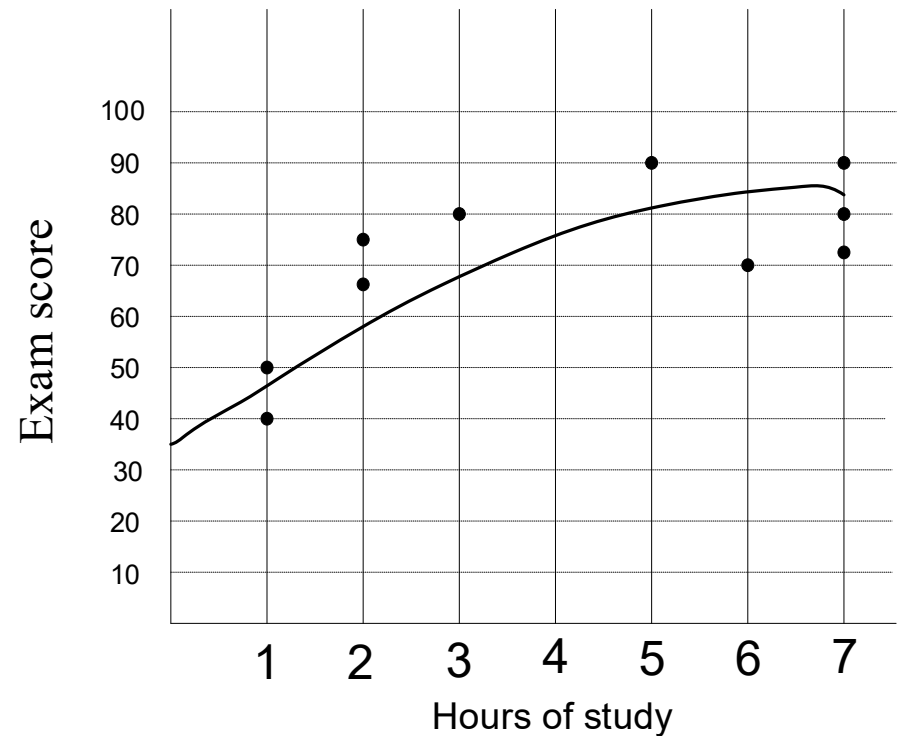
- In 1973, a famous statistician, Francis Anscombe, demonstrated how important it is to visualize the data. The concept got extended later to create [Datasaurus Dozen](https://cran.r-project.org/web/packages/datasauRus/vignettes/Datasaurus.html).
- It is a collection of 12 scatterplots with the same means, standard deviations, and correlation coefficient for X and Y (up to 2 decimal places).
- However, the shape of the data is very different from each other. Therefore, the scatterplots tell very different stories about the behavior and interrelationships of X and Y.
- Data available at <https://cran.r-project.org/web/packages/datasauRus/vignettes/Datasaurus.html>

Correlation Analysis



We need to measure the degree of correlation between two attributes.

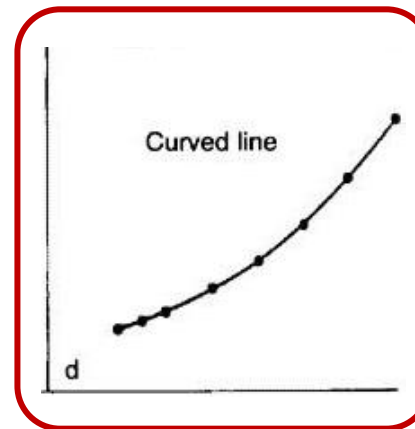
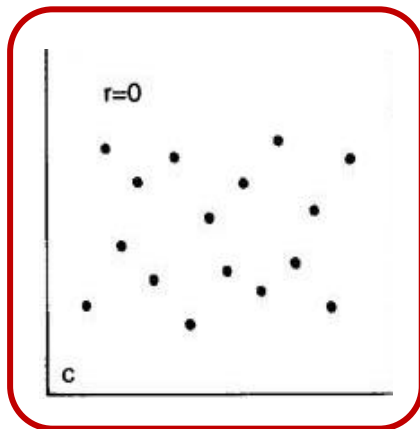
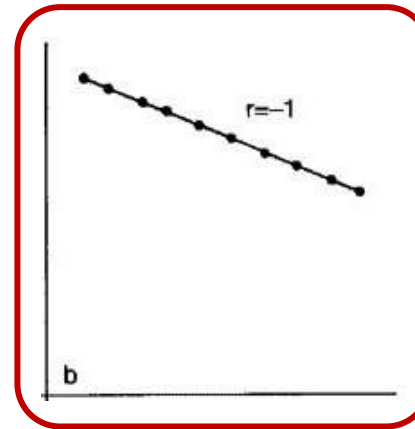
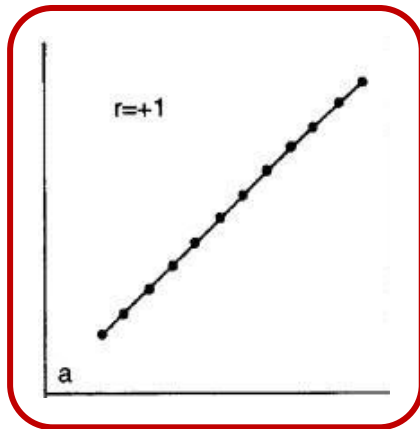
<i>Hours Study</i>	<i>Exam Score</i>
3	80
5	90
2	75
6	80
7	90
1	50
2	65
7	85
1	40
7	100



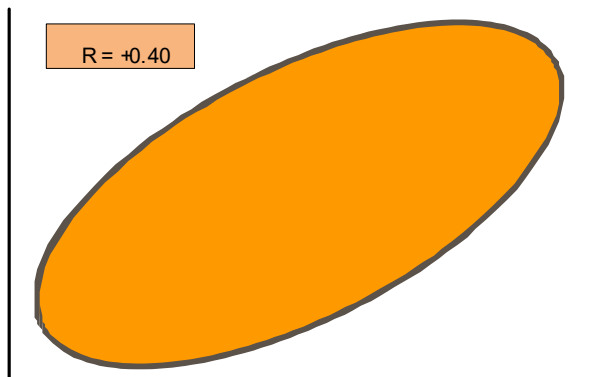
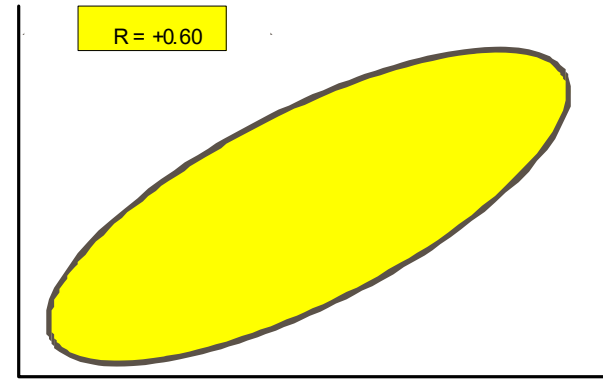
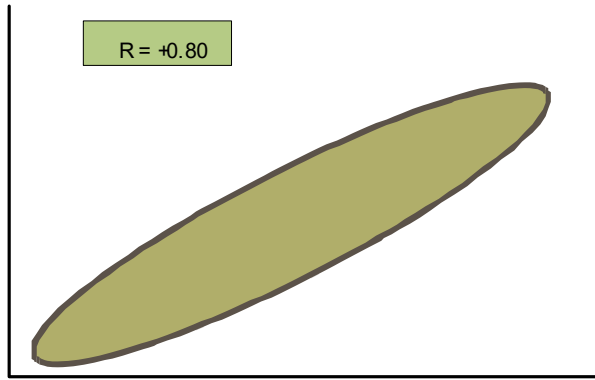
Correlation Coefficient

- 🌐 Correlation coefficient is used to measure the **degree of association**.
- 🌐 It is usually denoted by r .
- 🌐 The value of r lies between $+1$ and -1 .
- 🌐 Positive values of r indicates positive correlation between two variables, whereas, negative values of r indicate negative correlation.
- 🌐 $r = +1$ implies **perfect positive correlation**, and otherwise.
- 🌐 The value of r nearer to $+1$ or -1 indicates **high degree of correlation** between the two variables.
- 🌐 $r = 0$ implies, there is **no correlation**

Correlation Coefficient



Correlation Coefficient



Measuring Correlation Coefficients



Three methods to measure the correlation coefficients

Karl Pearson's coefficient

Find correlation coefficient between two **numerical** attributes

Charles Spearman's coefficient

Find correlation coefficient between two **ordinal** attributes

Chi-square coefficient of correlation

Find correlation coefficient between two **nominal** attributes



Pearson's Correlation Analysis

Karl Pearson's Correlation Analysis

i This is also called Pearson's Product Moment Correlation

Definition : Karl Pearson's correlation coefficient

Let us consider two attributes are X and Y .

The Karl Pearson's coefficient of correlation is denoted by r^* and is defined as

$$r^* = \frac{cov(X, Y)}{\sigma_x \sigma_y} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}}$$

where

X_i = i – th value of X – variable, $\bar{X}$ = mean of X

Y_i = i – th value of Y – variable, $\bar{Y}$ = mean of Y

n = number of pairs of observation of X and Y

$cov(X, Y)$ = covariance of X and Y , σ_X = SD of X , σ_Y = SD of Y

Covariance Vs. Correlation

Measures of the relationship between two variables

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

correlation between X and Y

standard deviation of X

standard deviation of Y

covariance normalized by standard deviation

Correlation

- measures the **strength** and **direction** of the linear relationship between two variables.
- standardized values
- not** affected by scale
- value range $[-1 \dots 1]$

$$\text{corr}_{XY} = \rho_{XY} = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sigma_x \sigma_y}$$

Expected value operator

Random variables means

Covariance

- measures the **extent** to which two variables are linearly related or dependent,
- not** standardized values
- affected by scale of the variable
- value range $[-\infty \dots \infty]$

$$\text{cov}_{XY} = \sigma_{XY} = E[(X - \mu_x)(Y - \mu_y)]$$

Correlation Coefficient

Since r is the ratio of the covariance to the product of the standard deviations:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

By the Cauchy-Schwarz inequality, we know:

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}$$

Therefore:

$$|r| = \left| \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}} \right| \leq 1$$

This inequality shows that the correlation coefficient r must lie between -1 and 1.

Karl Pearson's Coefficient of Correlation

Example : Correlation of Gestational Age and Birth Weight

A small study is conducted involving 17 infants to investigate the association between gestational age at birth, measured in weeks, and birth weight, measured in grams.

Infant ID #	Gestational Age (wks)	Birth Weight (gm)
1	34.7	1895
2	36.0	2030
3	29.3	1440
4	40.1	2835
5	35.7	3090
6	42.4	3827
7	40.3	3260
8	37.3	2690
9	40.9	3285
10	38.3	2920
11	38.5	3430
12	41.4	3657
13	39.7	3685
14	39.7	3345
15	41.1	3260
16	38.0	2680
17	38.7	2005

Karl Pearson's coefficient of Correlation



Example : Correlation of Gestational Age and Birth Weight

A small study is conducted involving 17 infants to investigate the association between gestational age at birth, measured in weeks, and birth weight, measured in grams.

We wish to estimate the association between gestational age and infant birth weight.

Birth weight → Dependent variable

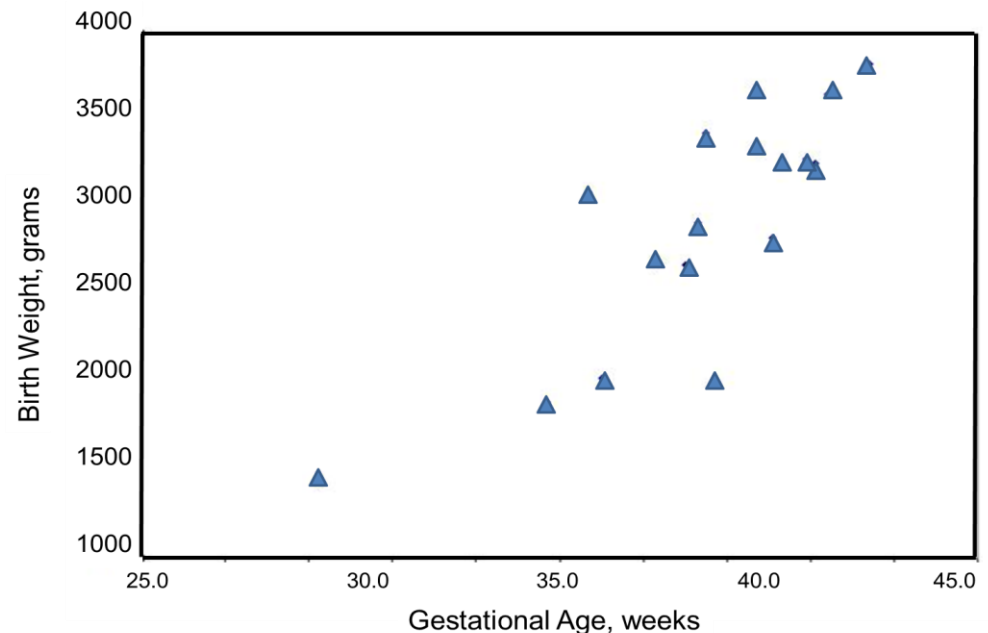
Gestational age → Independent variable

Thus

$Y = \text{birth weight}$ and

$X = \text{gestational age}$

The data are displayed in the scatter diagram.



Karl Pearson's coefficient of Correlation



Infant ID #	Gestational Age (wks)	Birth Weight (gm)
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11	38.5	3430
12	41.4	3657
13	39.7	3685
14	39.7	3345
15	41.1	3260
16	38.0	2680
17	38.7	2005

For the given data

$$\bar{X} = \frac{\sum X}{n} = \frac{652.1}{17} = 38.4$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{49334}{17} = 2902$$

$$S_x^2 = \frac{\sum (X - \bar{X})^2}{n-1} = \frac{159.45}{16} = 9.97$$

$$S_y^2 = \frac{\sum (Y - \bar{Y})^2}{n-1} = \frac{7767660}{16} = 485578.8$$

$$r^* = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}} = 0.82$$

Conclusion: The sample's correlation coefficient indicates a strong positive correlation between Gestational Age and Birth Weight.



Rank Correlation Analysis

Charles Spearman's Correlation Coefficient



This correlation measurement is also called Rank correlation

- This technique is applicable to determine the degree of correlation between two variables in case of **ordinal data**.
- We can assign rank to the different values of a variable with ordinal data type.

Example

Height: [VS S N T VT]

5 4 3 2 1

T – shirt: [XXL XL L S VS]

1 2 3 4 5

Rank assigned

Charles Spearman's Correlation Coefficient



Definition: Charles Spearman's correlation coefficient

The rank correlation can be defined as

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

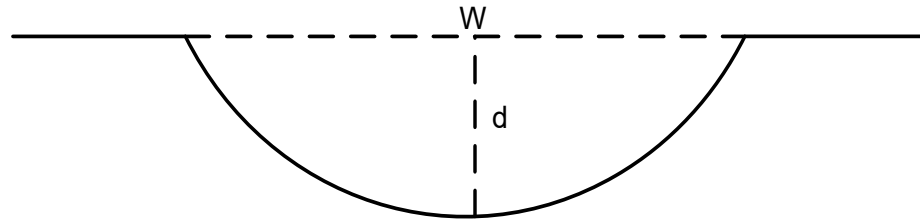
where d_i = Difference between ranks of i^{th} pair of the two variables
 n = Number of pairs of observations

- The Spearman's coefficient is often used as a statistical methods to aid either providing or disproving a hypothesis.

Charles Spearman's Coefficient of Correlation



Example: The hypothesis that the depth of a river **does not progressively increase** further from the bank.



A sample of size 10 is collected to test the hypothesis, using Spearman's correlation coefficient.

<i>Sample#</i>	<i>Width in m</i>	<i>Depth in m</i>
1	0	0
2	50	10
3	150	28
4	200	42
5	250	59
6	300	51
7	350	73
8	400	85
9	450	104
10	500	96



Charles Spearman's Coefficient of Correlation

Step 1: Assign rank to each data. It is customary to assign rank 1 to the largest data, and 2 to next largest and so on.

Note: If there are two or more samples with the same value, the mean rank should be used.

<i>Data</i>	20	25	25	25	30
<i>Assign rank</i>	5	4	3	2	1
<i>Final rank</i>	5	3	3	3	1



Charles Spearman's Coefficient of Correlation

Step 2: The contingency table will look like

<i>Sample#</i>	<i>Width</i>	<i>Width rank</i>	<i>Depth</i>	<i>Depth rank</i>	<i>d</i>	<i>d</i> ²
1	0	10	0	10	0	0
2	50	9	10	9	0	0
3	150	8	28	8	0	0
4	200	7	42	7	0	0
5	250	6	59	5	1	1
6	300	5	51	6	-1	1
7	350	4	73	4	0	0
8	400	3	85	3	0	0
9	450	2	104	1	1	1
10	500	1	96	2	-1	1

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 4}{10 \times 99}$$

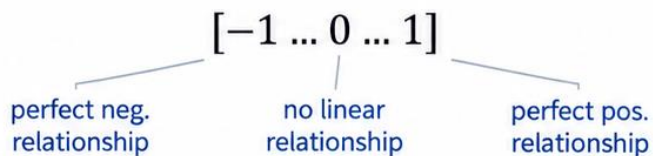
$$r_s = 0.9757$$

$$\sum d^2 = 4$$

Pearson Vs. Spearman Correlation

Pearson correlation coeff.

measures strength of **linear** relationship between two variables

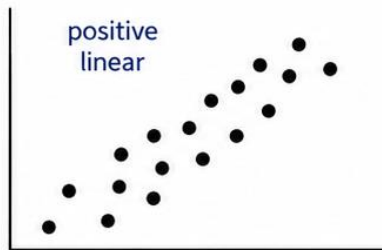


Assumptions

- normally distributed data
- linear relationship between variables

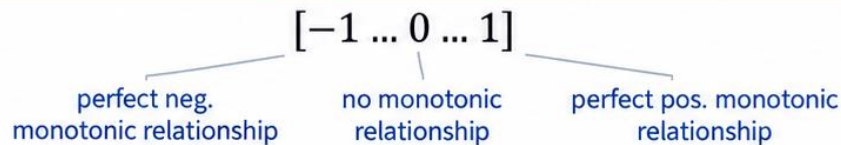
- Sensitivity to outliers
- Requires larger size than Spearman due to normality assumption
- Best suited for continuous data

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum [(X_i - \mu_X) \cdot (Y_i - \mu_Y)]}{n - 1}$$



Spearman correlation coeff.

measures strength of monotonic relationship between two variables, **whether it is linear or not**



Assumptions

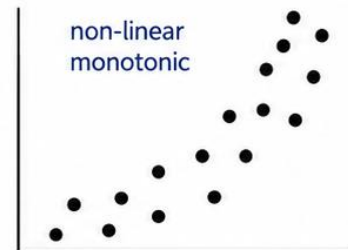
- no assumption on data distribution

- Can be used with any type of data (including ordinal)

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$d_i = R(X_i) - R(Y_i)$
 raw score
 rank (positions)
 of score

num.
observations





χ^2 Correlation Analysis

Chi-Squared Test of Correlation

- This method is also alternatively termed as **Pearson's χ^2 -test** or simply **χ^2 -test**
- This method is applicable to categorical (discrete) data only.

- Suppose, two attributes A and B with categorical values

$$A = a_1, a_2, a_3, \dots, a_m \quad \text{and}$$

$$B = b_1, b_2, b_3, \dots, b_n$$

having m and n distinct values.

A	a_1	a_2	a_3	a_1	a_5	a_1	
B	b_1	b_2	b_3	b_1	b_5	b_1	

Between whom we are to find the correlation relationship.

χ^2 –Test Methodology



Contingency Table

Given a data set, it is customary to draw a contingency table, whose structure is given below.

	b_1	b_2	-----	b_j	-----	b_n	Row Total
a_1							
a_2							
⋮							
a_i							
⋮							
a_m							
Column Total							Grand Total

χ^2 –Test Methodology



Entry into Contingency Table: Observed Frequency

In contingency table, an entry O_{ij} denotes the event that attribute A takes on value a_i and attribute B takes on value b_j (i.e., $A = a_i, B = b_j$).

A	a_1	a_2	a_3	a_i	a_5	a_i	
B	b_j	b_2	b_3	b_j	b_5	b_j	

	b_1	b_2	-----	b_j	-----	b_n	Row Total
a_1							
a_2							
$\vdots$							
a_i				O_{ij}			
$\vdots$							
a_m							
Column Total							Grand Total

χ^2 –Test Methodology



Entry into Contingency Table: **Expected Frequency**

In contingency table, an entry e_{ij} denotes the expected frequency, which can be calculated as

$$e_{ij} = \frac{\text{Count}(A = a_i) \times \text{Count}(B = b_j)}{\text{Grand Total}} = \frac{A_i \times B_j}{N}$$

	b_1	b_2	-----	b_j	-----	b_n	Row Total
a_1							
a_2							
$\vdots$							
a_i				e_{ij}			A_i
$\vdots$							
a_m							
Column Total				B_j			N

A	B
...	...
a_i	b_j
...	...
a_i	b_j
...	...
...	...
a_i	b_j
...	...
...	...
...	...

χ^2 – Test

Definition: χ^2 -Value

The χ^2 value (also known as the Pearson's χ^2 test) can be computes as

$$\chi^2 = \sum_{i=1}^m \sum_{j=1}^n \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the observed frequency
 e_{ij} is the expected frequency

χ^2 – Test



- The cell that contribute the most to the χ^2 value are those whose actual count is very different from the expected.
- The χ^2 statistics tests the hypothesis that A and B are independent. The test is based on a significance level, with $(n-1) \times (m-1)$ degrees of freedom., with a contingency table of size $n \times m$
- If the hypothesis can be rejected, then we say that A and B are statistically related or associated.

χ^2 – Test



Example 3: Survey on Gender versus Hobby.

- Suppose, a survey was conducted among a population of size 1500. In this survey, gender of each person and their hobby as either “book” or “computer” was noted. The survey result obtained in a table like the following.

GENDER	HOBBY
*****	*****
*****	*****
M	Book
F	Computer
*****	*****
*****	*****
*****	*****

- We have to find if there is any association between Gender and Hobby of a people, that is, we are to test whether “gender” and “hobby” are correlated.

χ^2 – Test



Example : Survey on Gender versus Hobby.

From the survey table, the **observed frequency** are counted and entered into the contingency table, which is shown below.

GENDER	HOBBY
.....	
.....	
M	Book
F	Computer
.....	



HOBBY	GENDER			
		Male	Female	Total
	Book			
	Computer			
	Total			

χ^2 – Test



Example: Survey on Gender versus Hobby.

- From the survey table, the observed frequency are counted and entered into the contingency table, which is shown below.

		GENDER		
		Male	Female	Total
HOBBY	Book	250	200	450
	Computer	50	1000	1050
	Total	300	1200	1500

χ^2 – Test



Example: Survey on Gender versus Hobby.

- From the survey table, the **expected frequency** are counted and entered into the contingency table, which is shown below.

HOBBY	GENDER		
		Male	Female
HOBBY	Book	90	360
	Computer	210	840
	Total	300	1200

χ^2 – Test

- Using equation for χ^2 computation, we get

$$\begin{aligned}\chi^2 &= \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} \\ &= 507.93\end{aligned}$$

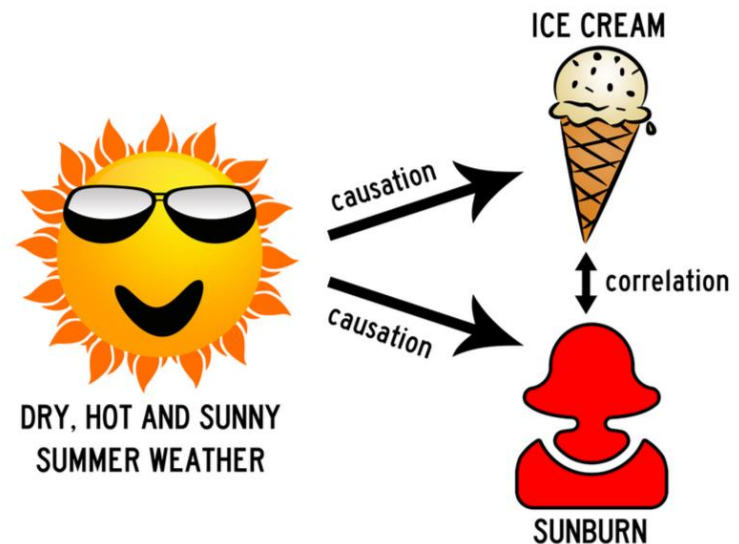
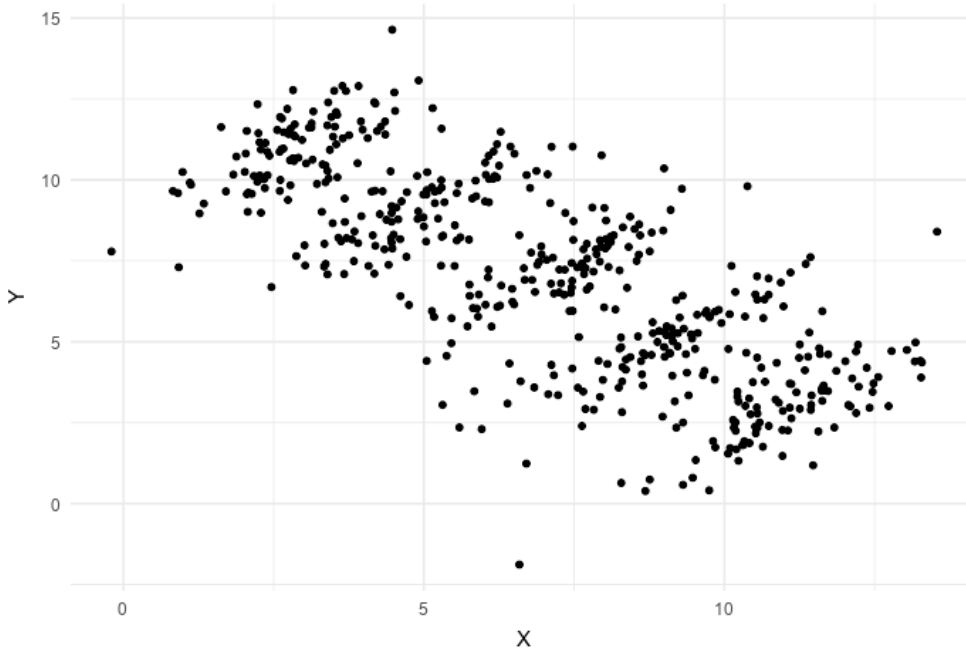
- This value needs to be compared with the tabulated value of χ^2 (available in any standard book on statistics) with 1 degree of freedom (for a table of $m \times n$, the degrees of freedom is $(m - 1) \times (n - 1)$; here $m = 2$, $n = 2$).
- For 1 degree of freedom, the χ^2 value needed to reject the hypothesis at the 0.01 significance level is 10.828. Since our computed value is above this, we reject the hypothesis that “Gender” and “Hobby” are independent and hence, conclude that the two attributes are *strongly correlated* for the given group of people.



Correlation does NOT imply Causation

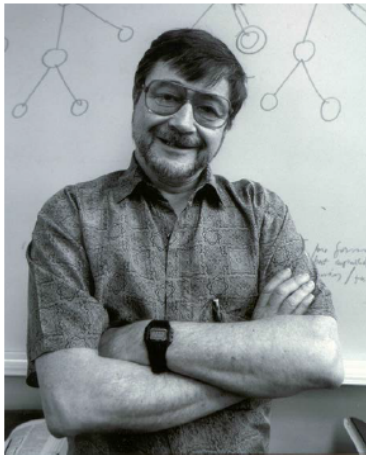
Correlation does not imply causation

Korrelation:



A Paradigm Shift: Causality

- **Causality theory** helps to decide when, and how, causation can be inferred from domain knowledge and data.
- “...all approaches to causation are variants or abstractions of structural theory.” - Judea Pearl.
- The modeling of the underlying structures provides a language to encode causal relationships – the basis of a causality theory.



Judea Pearl
(*1936)

ACM Turing Award 2011:

"For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning."

